Iterative Prompting

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

THE CONNECTED WEEK

Teach iteration as recorded diagnosis, not repeated guessing.

Supply exact source/prompt/output cards, keep R unrun and assess the selected format-revision plan.

Keep one habit: Ask which failure motivated the change, which conditions stayed fixed and which outcomes were actually supplied.

MAIN LESSON · SECTION 1

Prepare this connected week

> Use one main plan and keep practice separate from the independent first attempt.

Learning goal and precision

Primary target: Explain the **mechanism** and terminology of controlled prompt experiments

Two 45-minute sessions. A prompt experiment compares complete specifications using the same task cases. Define success before inspecting **output**s; change one intended instruction and preserve other settings.

Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions.

Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

By the end, I can revise one prompt component and compare paired **output**s using the same **rubric**.

What to prepare

- Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.
- Case exposure check: this source example appears in ai-2/28, ai-3/7, ai-3/9. Reuse a familiar case for recall or modelling; verify that the case used to assess transfer has not already received feedback.
- Lesson and workbook; paper and pencil
- Calculator and optional local Python 3 for arithmetic; no live API or private data
- Prepare the fresh retry separately and withhold its solution until after collection

Main teaching plan

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Opening retrieval and readiness

Without opening the old lesson, explain one key idea from Examples and **Rubric**s. Give an example and a mistake that the idea helps you avoid.

Look for: Distinguish a format example from approved factual evidence

Reference example, not the only acceptable wording: A format example demonstrates structure; approved N supplies the frog fact. The original mammal answer fails mandatory support even with3/4points, while the answer-only repair still lacks the Question field. Check each required criterion separately.

Without opening the old lesson, explain one key idea from Sampling and Variability. Give an example and a mistake that the idea helps you avoid.

Look for: Sample from a distribution using explicit cumulative intervals

Reference example, not the only acceptable wording: Teacher explanation: Selections red,blue,green,green. The draw 1 violates the half-open [0,1) **input** contract and must be rejected.

Without opening the old lesson, explain one key idea from Predictive and Generative AI. Give an example and a mistake that the idea helps you avoid.

Look for: Distinguish discriminative **prediction** from generation by inspecting the requested **output**

Reference example, not the only acceptable wording: Teacher explanation: Message labels are discriminative **prediction**; continuing a sequence is generation; record lookup is retrieval rather than learned generation by itself.

One application can route a request, retrieve **evidence** and generate a response.

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

▼ **Earlier wording, alternatives and picture transcripts**

Teach the learning cycle

Retrieval answer guidance

Examples and Rubrics

Sampling and Variability

Predictive and Generative AI

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

A summary has accurate facts but uses unexplained terms. Propose one prompt change and a comparison that checks both accessibility and factual preservation.

Concept to explain

Revision should respond to diagnosed **output** failures.

Term	Meaning
iteration	One repeat of a process, often followed by inspection and revision.
failure diagnosis	An investigation locating where and why a process failed.
version	An identified state of a document, dataset, model or program.
change log	A record of what changed, when it changed and why.

Prepare the class

This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Readiness answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Model a focused revision and its evidence limits

> Keep the six-case quantitative review separate from the four-question prompt audit.

State the controlled-comparison contract

A prompt experiment compares complete specifications using the same task cases. Define success before inspecting **output**s; change one intended instruction and preserve other settings.

Evaluate both prompts on **paired case**s with fixed model/**version**, source records, **output** contract and scoring **rubric**. Use development results for selection, then reserve new final cases.

Score each case rather than counting attractive sentences.

Trace the separate six-case record

Separate six-case development record	P	Q
Passing IDs	1,2,3,4	1,2,3,5,6
Missing-approval slice IDs4,6	passes4, fails6	fails4, passes6
All-case denominator	6	6

On six development cases P passes cases 1, 2, 3, 4; Q passes 1, 2, 3, 5, 6. Missing-approval cases are 4 and 6. Compare totals and condition results.

P passes 4/6 and Q 5/6. P passes one of two missing-approval cases, Q also one of two. Q improves the aggregate while leaving this condition unchanged.

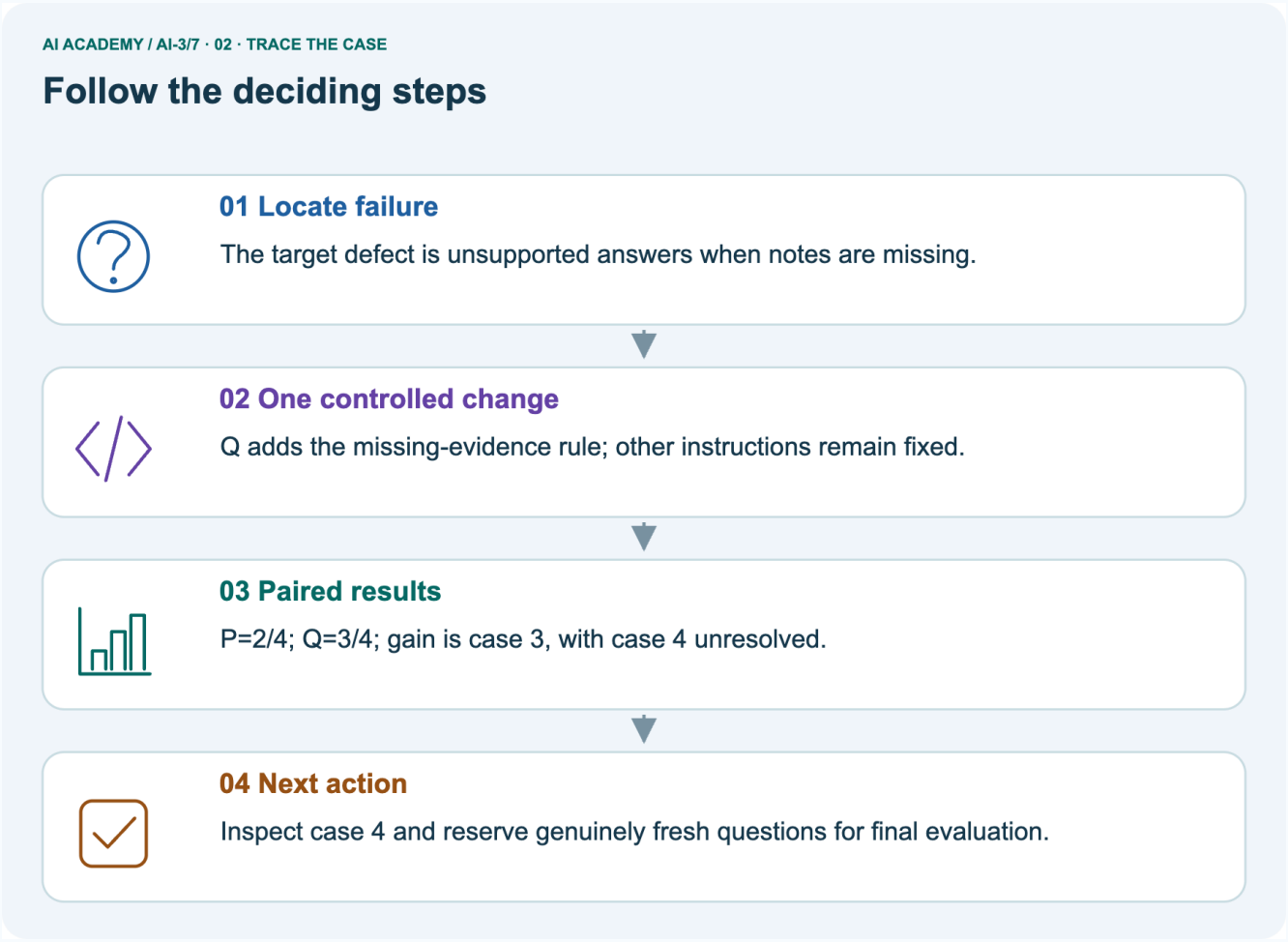
Read the pictured missing-definition change

A fictional Study Buddy prompt invents a definition when approved notes omit it.

P asks “answer helpfully”. Revision Q adds “if approved notes lack the answer, say not supplied”.

Hold model settings and four development questions fixed. P passes 1,2; Q passes 1,2,3. Case 4 still fails.

Trace the case3 gain and case4 failure



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Locate failure	The target defect is unsupported answers when notes are missing.
2. One controlled change	Q adds the missing-evidence rule; other instructions remain fixed.
3. Paired results	P=2/4; Q=3/4; gain is case 3, with case 4 unresolved.
4. Next action	Inspect case 4 and reserve genuinely fresh questions for final evaluation.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

An **iteration** should respond to a recorded failure with a precise change. Comparing the same questions reveals whether other behavior changed too.

The invented results show a development gain of one case, not a guarantee that the new instruction always works. Save prompt text, settings, source notes and case-level outcomes.

Reject simultaneous source/settings changes

Change: Change model settings and note wording at the same time as the prompt.

Prediction: The comparison no longer isolates the prompt revision.

Reason: Several changes could explain the altered **output**s; record them or run separate controlled comparisons.

Explain small-run and final-evaluation limits

Stochastic **output** can vary across runs; a small comparison does not prove an average effect or classroom learning benefit.

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Explain the mechanism

Input → **Process** → **Output**

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

1. **Version** 1 fails: answer is too advanced.

2. Change only audience constraint; **Version** 2 is age-fit but lacks **evidence**.

3. Add source-use rule; **Version** 3 cites the supplied passage.

4. Record scores and reasons so improvement is traceable.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

A factually correct explanation is too difficult for a nine-year-old.

Make and test one focused revision.
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Exact explanation to model

Constrain vocabulary and request a concrete example. Compare readability and factual preservation on the same passages.

Explain the added worked example

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Key terms

Term	Meaning
Paired case	The same case evaluated under each compared procedure
Rubric	Prespecified criteria for judging an output
Regression	A previously passing condition that fails after a change

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

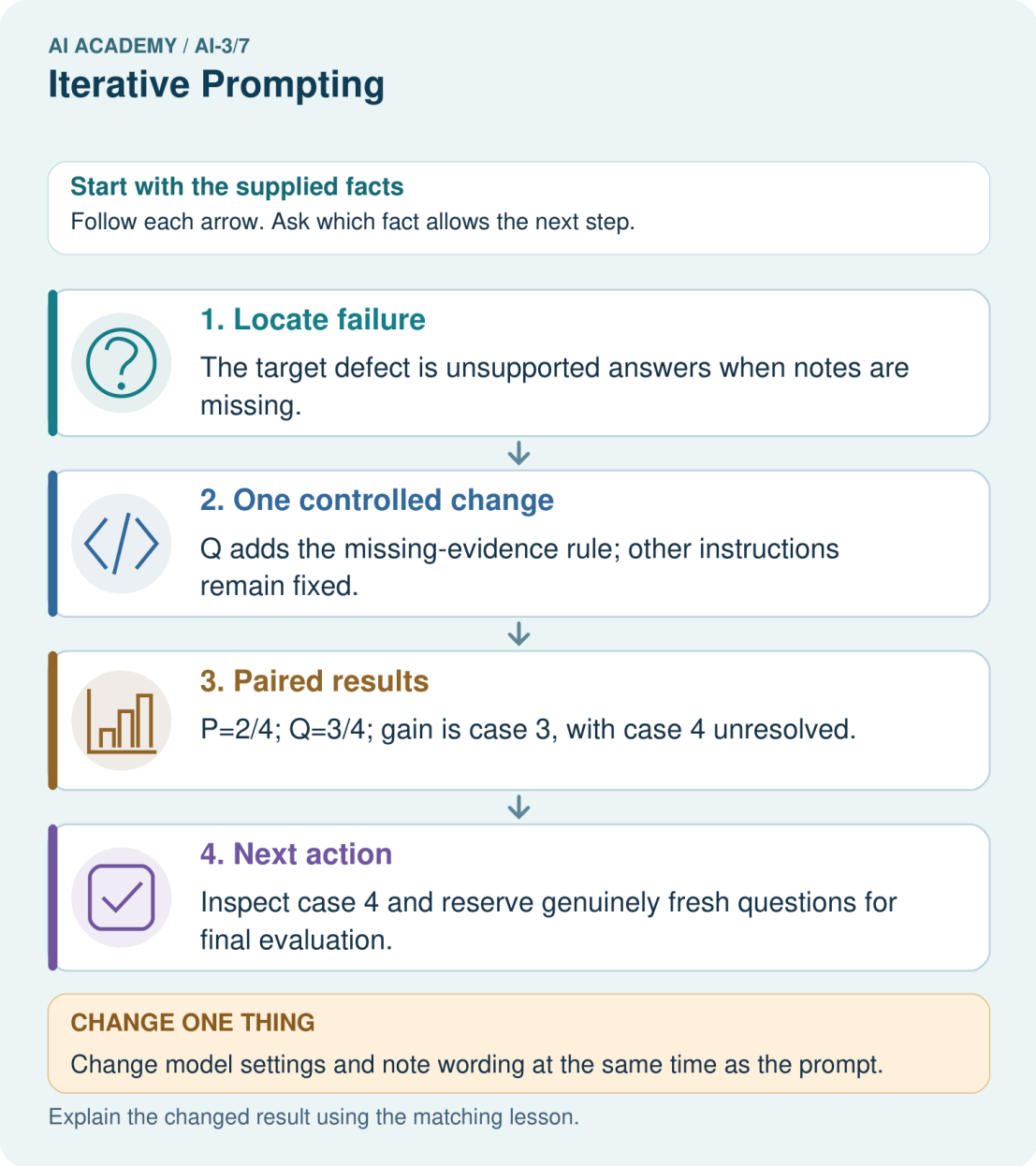
The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Iterative Prompting

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

MAIN LESSON · SECTION 3

Guide the explanation, then remove help

> Collect each learner’s reasoning before showing feedback.

Main completion prompt

Completion task: On six development cases P passes cases 1, 2, 3, 4; Q passes 1, 2, 3, 5, 6. Missing-approval cases are 4 and 6. Compare totals and condition results.

Show the intermediate steps and give the conclusion with its scope.

Expected reasoning and fading help

Teacher check: Teacher explanation: P passes 4/6 and Q 5/6. P passes one of two missing-approval cases, Q also one of two. Q improves the aggregate while leaving this condition unchanged.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for Controlled prompt experiments. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Prompt selection on development cases cannot be reported as independent final performance.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial question and changed-case reasoning

What **evidence** supports improvement here, and what remains unproven?

Q gains case 3 without losing 1 or 2 on this four-case record; 4 still fails and new-case performance remains unknown.

Changed-case reasoning: The comparison no longer isolates the prompt revision. Several changes could explain the altered **output**s; record them or run separate controlled comparisons.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Keep each learner explaining

- Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.
- Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.
- In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

▼ Earlier wording, alternatives and picture transcripts

Demonstrate and guide practice

Board sequence

Write the given facts: A factually correct explanation is too difficult for a nine-year-old.

Write the question to explain: Make and test one focused revision.

Reveal after **prediction**s: Constrain vocabulary and request a concrete example. Compare readability and factual preservation on the same passages.

Use the earlier changed question as optional later practice. Reserve Design a new format-revision evaluation independently for the first independent attempt before hints or feedback.

Classroom dialogue

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

Model these guided questions

Guided: P passes 4/6 and Q 5/6. P passes one of two missing-approval cases, Q also one of two. Q improves the aggregate while leaving this condition unchanged.

Qualification: Prompt selection on development cases cannot be reported as independent final performance.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

Case-specific teaching sequence

Stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain section 1 and vocabulary. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.

Stage	Teacher action
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the supplied investigation

> Keep expected results, observed paper traces and remaining limits distinguishable.

Supplied materials and answer guidance

P/Q lab: condensation/freezing pass both; melting failsP but passesQ with not supplied; evaporation failsboth under N-only rubric. Totals2/4→3/4, one gain,no losses,one unresolved case. R adds a checking step but has no supplied outputs. Separate six-case review4/6→5/6 with missing-approval1/2both,including loss4 and gain6.

Keep the selected three-bullet evaluation design independent; the older eight-case arithmetic task is additional practice.

Run and record

1. Predict → Test → Record → Explain

2. Keep expected and actual results separate. Surprises are useful data when recorded honestly.

3. Use the complete source N, fixed task/rubric, P/Q output cards and three-**version** prompt log. Score every supplied result, preserve case4, and label R as a proposed unrun revision. No actual model calls or measured third-**version** improvement are supplied.

4. What did the hardest test teach you that the normal test did not?

Keep an inspectable record

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose the first misunderstanding

The worked result establishes every future case: Prompt selection on development cases cannot be reported as independent final performance.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Choose support from the response

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ Earlier wording, alternatives and picture transcripts

Run the weekly evidence lab

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Make the lab runnable

Use the supplied investigation for shared practice. Keep predictions and observed paper-trace results separate. Reserve Design a new format-revision evaluation independently for the individual first attempt before feedback; do not use the reserved case as a paired lab task.

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the selected new case

> Keep this key until after the first attempt; accept valid alternatives with defensible reasoning.

Selected independent question and key

Selected case: A factual response uses four paragraphs when the contract requires three bullets. Design a controlled prompt revision and evaluation.

Change the format instruction while keeping approved sources, tasks and evaluation conditions fixed. Compare format compliance and factual support.

Preserve separate fresh final cases; an attractive revision is not **evidence** that facts remain correct.

Judge explanation, trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair action

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Explain the mechanism

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher explanation: A prompt experiment compares complete specifications using the same task cases. Define success before inspecting **output**s; change one intended instruction and preserve other settings.

Evaluate both prompts on **paired case**s with fixed model/**version**, source records, **output** contract and scoring **rubric**. Use development results for selection, then reserve new final cases.

Score each case rather than counting attractive sentences.

W2 Trace the worked case

On six development cases P passes cases 1, 2, 3, 4; Q passes 1, 2, 3, 5, 6. Missing-approval cases are 4 and 6. Compare totals and condition results. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: P passes 4/6 and Q 5/6. P passes one of two missing-approval cases, Q also one of two. Q improves the aggregate while leaving this condition unchanged.

W3 Audit the limit

A report omits this qualification: Prompt selection on development cases cannot be reported as independent final performance. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: Prompt selection on development cases cannot be reported as independent final performance.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. Prompt selection on development cases cannot be reported as independent final performance.

The new case should test this particular **mechanism** rather than merely restate the original result.

▼ Earlier wording, alternatives and picture transcripts

Assess and explain the answers

Worked case answer

Constrain vocabulary and request a concrete example. Compare readability and factual preservation on the same passages.

Additional transfer practice question

You change a prompt's audience, length and source rules together; the result improves. What can you conclude, and how would you investigate the cause?

Reasoned practice answer

The combined revision improved this result; it does not identify which change helped. Compare one controlled change at a time on the same tasks.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “Three simultaneous edits reveal which single edit caused improvement.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Ask for plain Grade 6 wording with short definitions for unavoidable terms. Compare both **version**s on the same passages using readability and claim-support criteria.

Do not simplify by deleting essential qualifications or adding unsupported analogies.

Explanations for the additional workbook examples

W1 Explain the mechanism

W2 Trace the worked case

W3 Audit the limit

W4 Apply a new case independently

On eight **paired case**s A passes 1–5; B passes 1–4, 6, 7. Special cases are 5 and 8. Compare overall and special-case results; select by total passes.

Teacher explanation: A=5/8, B=6/8, so select B on development. Special-case passes A=1/2, B=0/2; the aggregate selection hides a **regression** requiring review.

W5 Design a revealing follow-up

Assess independent transfer

Selected case answer

Feedback that the learner uses

Connect, adapt and return

> Retain the project evidence and revisit the reasoning on a changed case.

Project evidence to keep

Maintain prompt **version**, diagnosed failure, change, and before/after **rubric** scores.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Maintain a Study Buddy experiment log linking failure, single revision and paired **rubric** results.

Support while preserving the reasoning goal

Provide one short approved passage; separate instruction, **input**, **output** and **evidence**. Stretch: introduce an unsupported claim or conflicting source and validate claim by claim with a fallback.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Selected independent task: change only the format instruction to require three bullets; hold sources,tasks,settings and factual criteria fixed. Evaluate format and claim support separately, retain failures and reserve fresh final cases. No revised outputs are given. Fresh retry: a source-only answer exceeds a40-word limit. Propose one length-instruction revision and a controlled comparison that checks both length and factual support without inventing results. The older eight-case rate task remains additional practice.

Plan the delayed check

Suggested return points in this level: ai-3/8, ai-3/10, ai-3/13. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Bridge to next week

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Next: Week 8: Structured **Output** and Schemas. Retrieve today's **evidence** and identify what carries forward and what is new.

Next is Week8: Structured Output and Schemas. Carry forward exact response criteria and failure records when checking syntax, schema and factual support separately.

▼ Earlier wording, alternatives and picture transcripts

Connect and extend learning

Verified Study Buddy

Support and stretch

Exit reflection

Bridge to the next week

Support and fresh reassessment

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference: alternative planning checklist**

REFERENCE · SECTION 7

Reference: alternative planning checklist

> Use as a reminder alongside the main route rather than teaching the cycle twice.

Preparation and schedule

Teaching target: Explain the **mechanism** and terminology of controlled prompt experiments

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
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> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
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>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

Practice and assessment reminders

Before class, solve the supported practice: On six development cases P passes cases 1, 2, 3, 4; Q passes 1, 2, 3, 5, 6. Missing-approval cases are 4 and 6. Compare totals and condition results.

Show the intermediate steps and give the conclusion with its scope.

Reasoned practice answer: Teacher explanation: P passes 4/6 and Q 5/6. P passes one of two missing-approval cases, Q also one of two. Q improves the aggregate while leaving this condition unchanged.

Check every learner before discussion: Use the worked case for Controlled prompt experiments. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Prompt selection on development cases cannot be reported as independent final performance.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Change model settings and note wording at the same time as the prompt. Require a **prediction** and a reason before comparing. Expected result: The comparison no longer isolates the prompt revision.

Reason: Several changes could explain the altered **output**s; record them or run separate controlled comparisons.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Make a transparent first **version**. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference: feedback and records**

REFERENCE · SECTION 8

Reference: feedback and records

> Use digital tools where they help retain an explanation and its evidence.

Keep practice and independent evidence separate

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Repair and portfolio records

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.
</details>

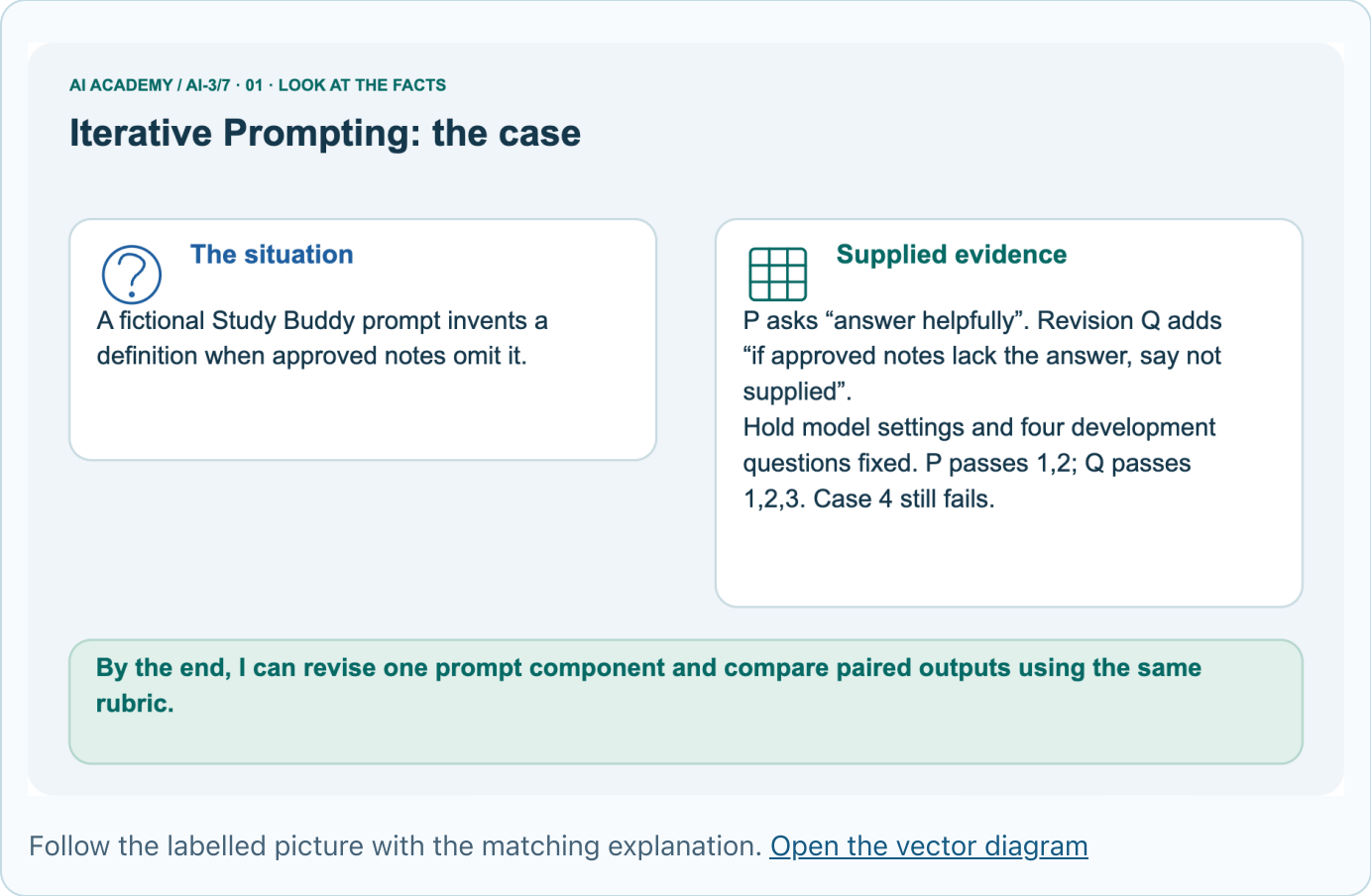
<details>
<summary>Reference: teach with the six pictures</summary>
</details>

REFERENCE · SECTION 9

Reference: teach with the six pictures

> Choose the picture for the teaching move and connect it to the explanation.

01. Iterative Prompting: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

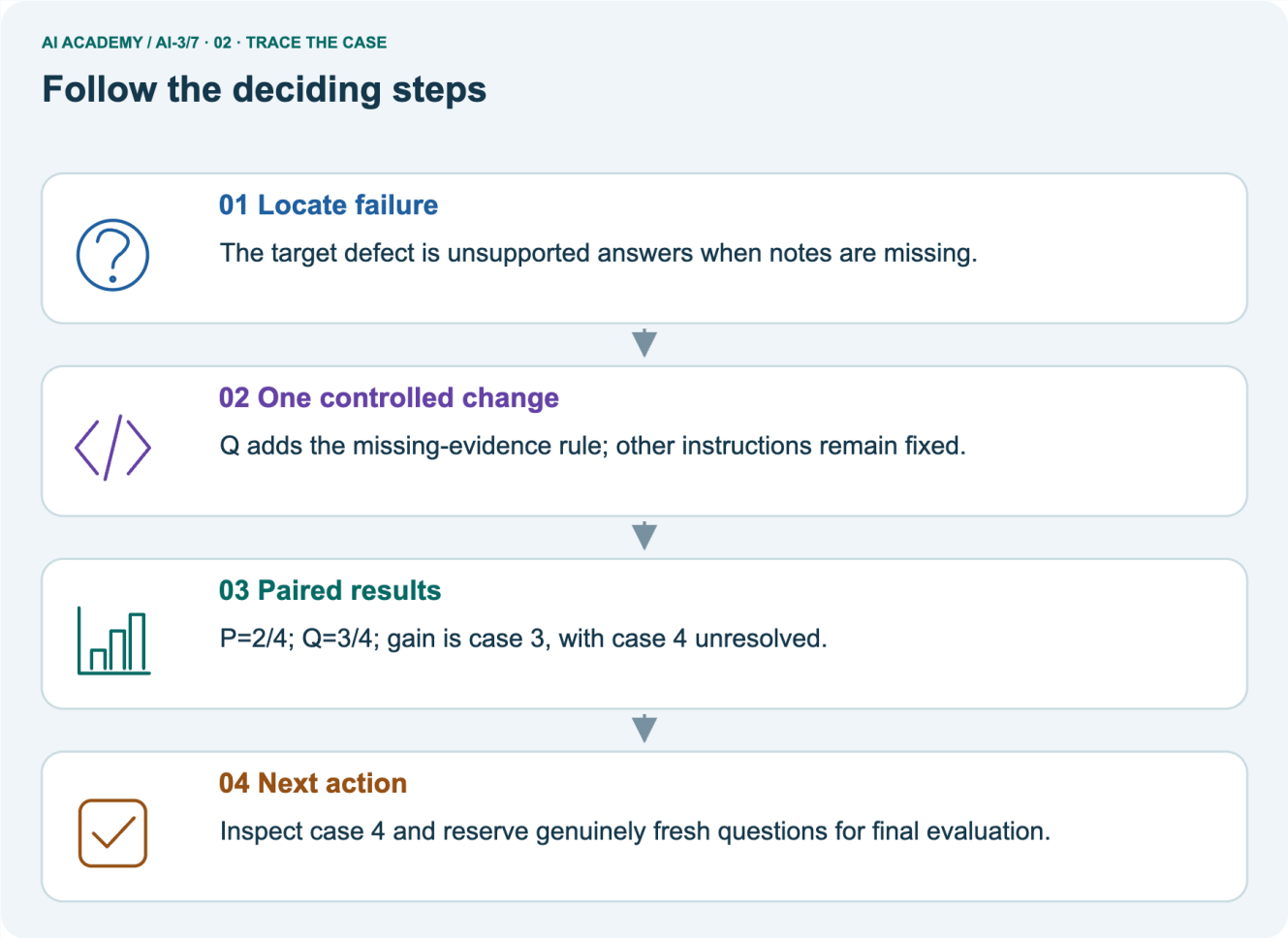
A fictional Study Buddy prompt invents a definition when approved notes omit it. Supplied facts: P asks “answer helpfully”. Revision Q adds “if approved notes lack the answer, say not supplied”.

Hold model settings and four development questions fixed. P passes 1,2; Q passes 1,2,3. Case 4 still fails.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



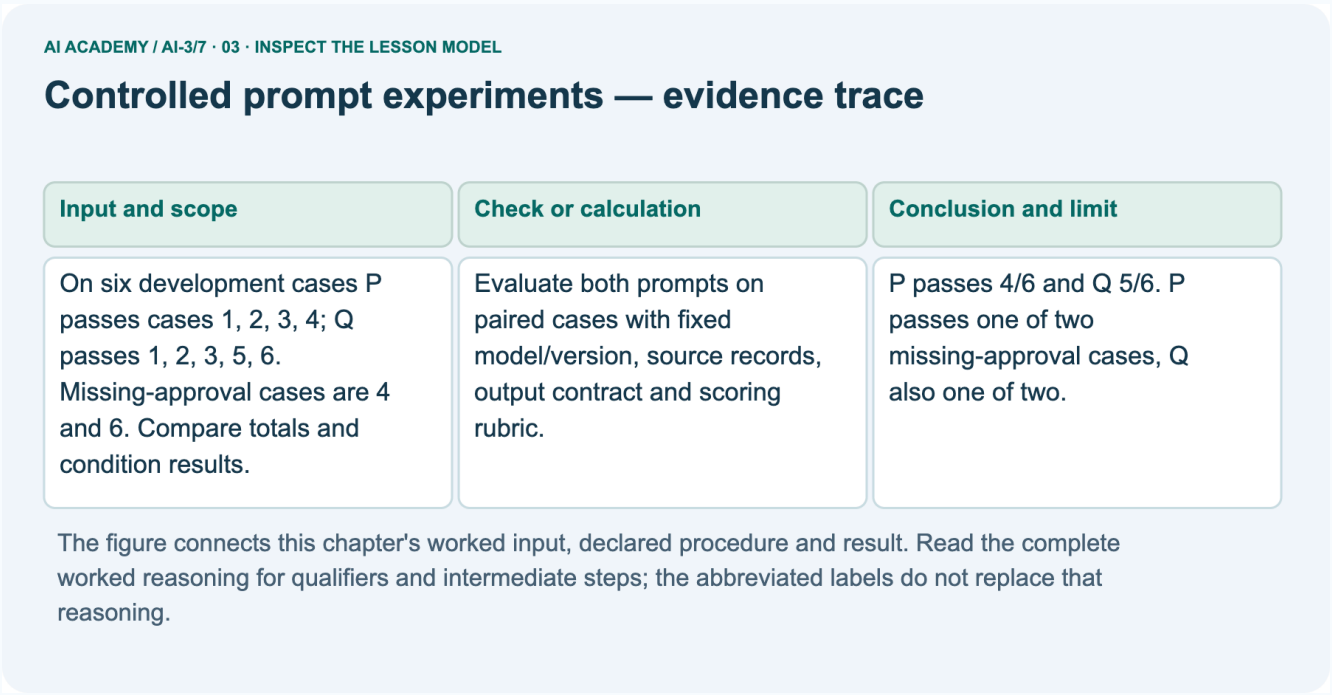
Follow the labelled picture with the matching explanation. Open the vector diagram

Locate failure: The target defect is unsupported answers when notes are missing. → One controlled change: Q adds the missing-**evidence** rule; other instructions remain fixed. → Paired results: P=2/4; Q=3/4; gain is case 3, with case 4 unresolved. → Next action: Inspect case 4 and reserve genuinely fresh questions for final evaluation.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Controlled prompt experiments — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Controlled prompt experiments — **evidence** trace. On six development cases P passes cases 1, 2, 3, 4; Q passes 1, 2, 3, 5, 6. Missing-approval cases are 4 and 6.

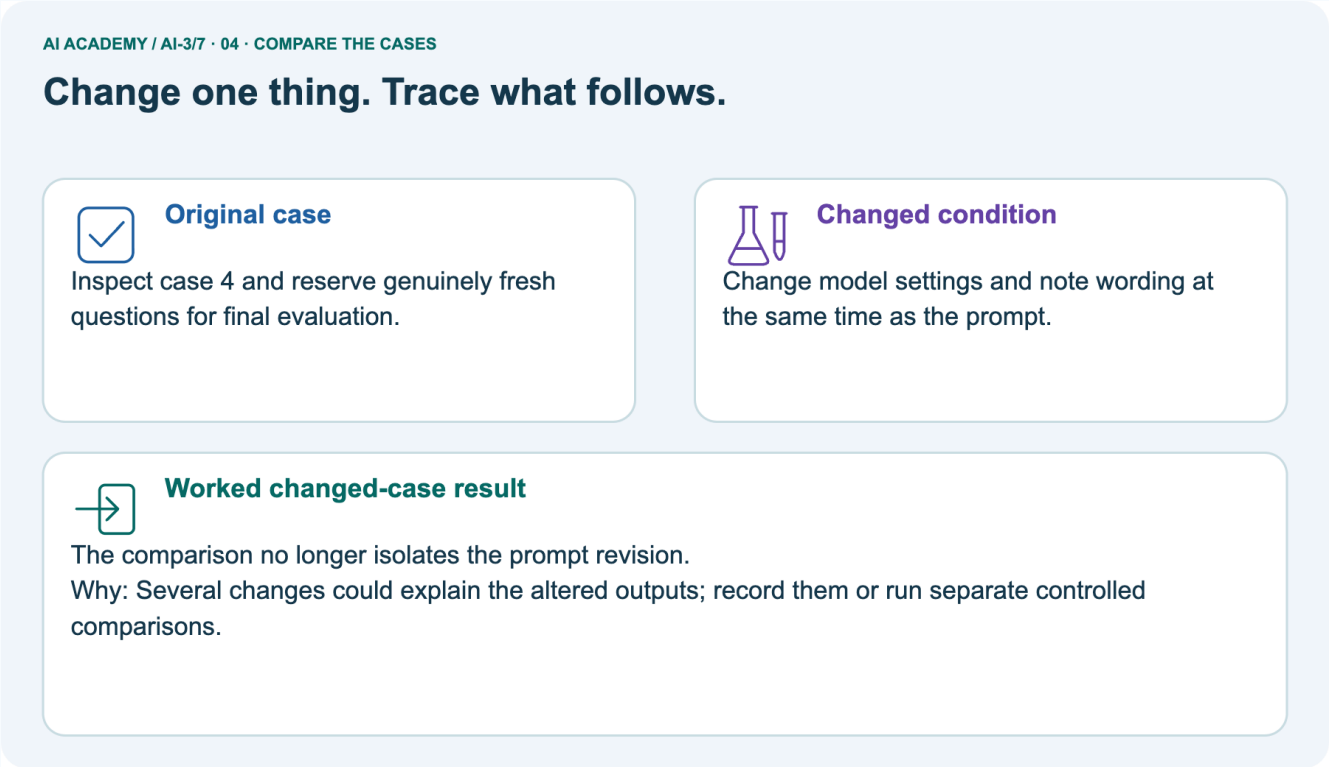
Compare totals and condition results. | Evaluate both prompts on **paired case**s with fixed model/**version**, source records, **output** contract and scoring **rubric**. | P passes 4/6 and Q 5/6.

P passes one of two missing-approval cases, Q also one of two.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Inspect case 4 and reserve genuinely fresh questions for final evaluation. Changed condition: Change model settings and note wording at the same time as the prompt.

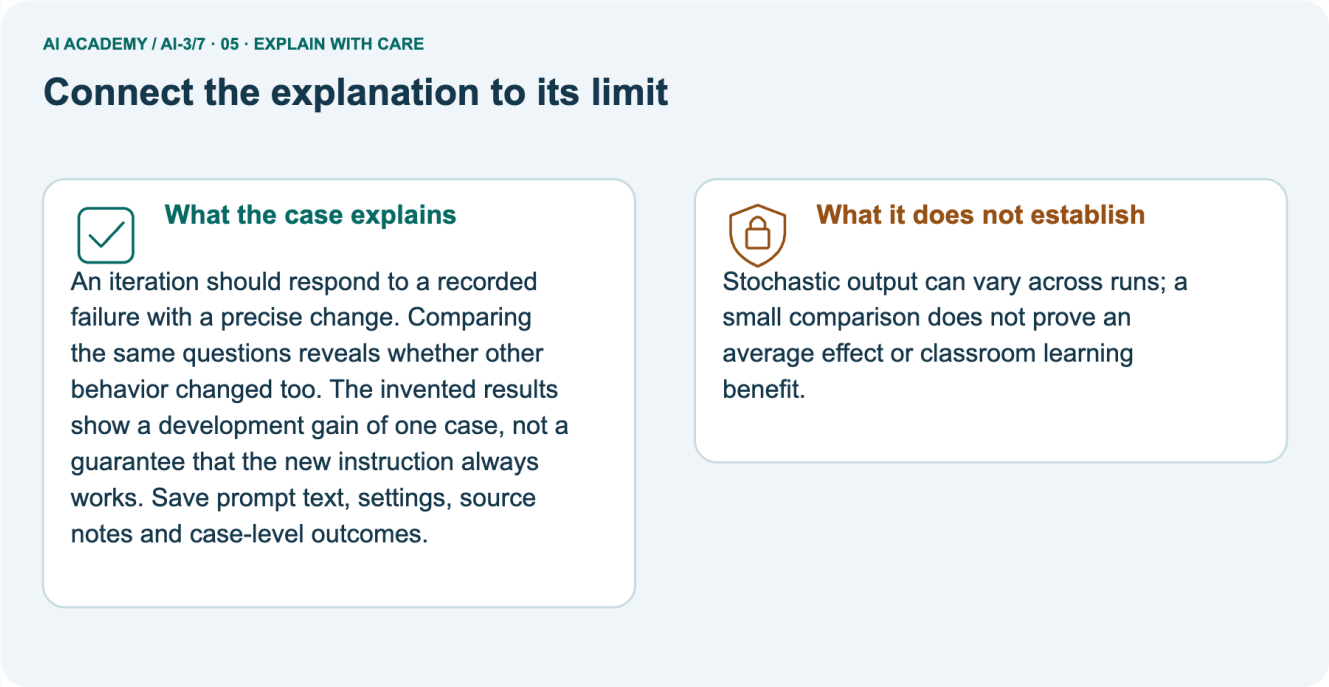
Expected result: The comparison no longer isolates the prompt revision. Reason: Several changes could explain the altered **output**s; record them or run separate controlled comparisons.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Several changes could explain the altered **output**s; record them or run separate controlled comparisons.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

An **iteration** should respond to a recorded failure with a precise change. Comparing the same questions reveals whether other behavior changed too.

The invented results show a development gain of one case, not a guarantee that the new instruction always works. Save prompt text, settings, source notes and case-level outcomes.

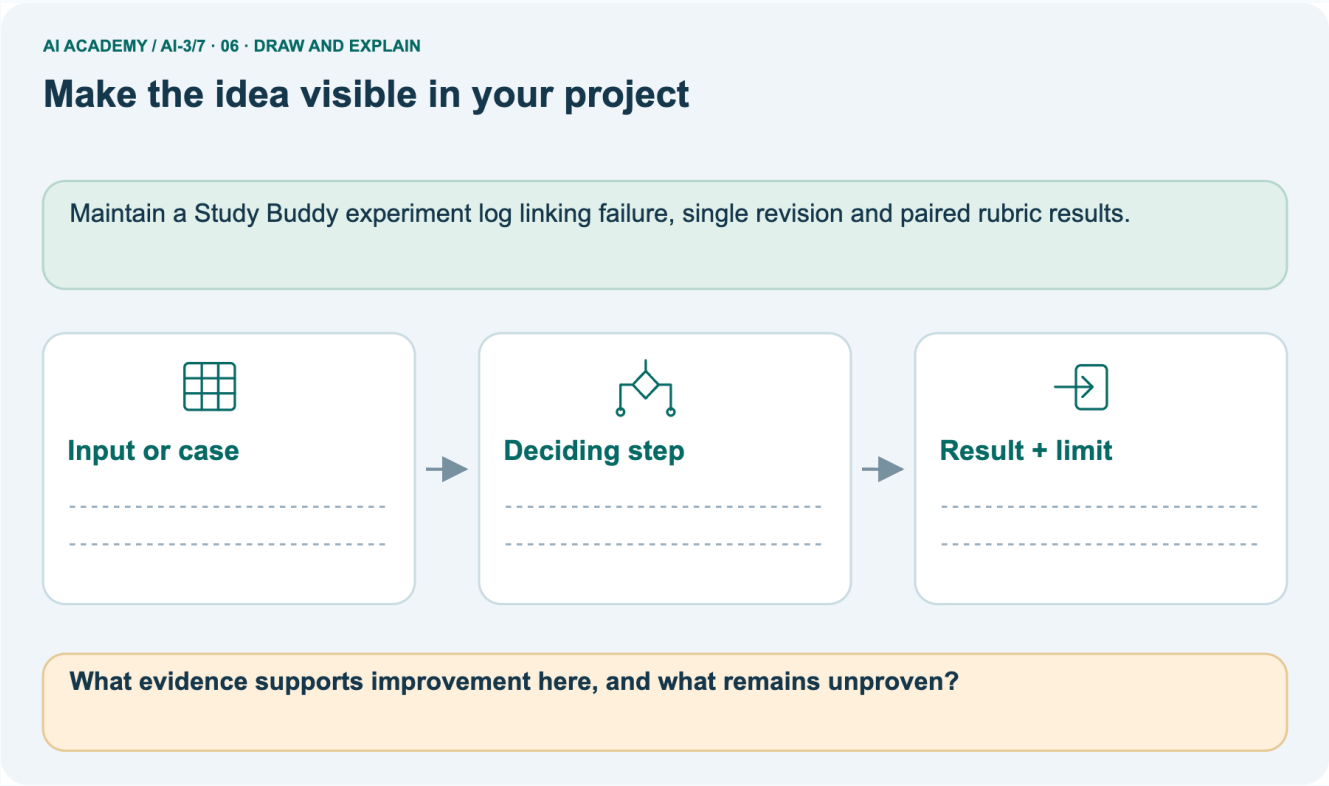
Limit: Stochastic **output** can vary across runs; a small comparison does not prove an average effect or classroom learning benefit.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Stochastic **output** can vary across runs; a small comparison does not prove an average effect or classroom learning benefit.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Maintain a Study Buddy experiment log linking failure, single revision and paired **rubric** results. Draw your **input**, deciding step, result and limit.

Supported question: What **evidence** supports improvement here, and what remains unproven?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Q gains case 3 without losing 1 or 2 on this four-case record; 4 still fails and new-case performance remains unknown.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Iterative Prompting: the case

02. Follow the deciding steps

03. Controlled prompt experiments — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.